

# Data Lifecycle Management Procedures V1.9

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| 1.0     | March 15, 2022    | Tariq Al-Rasheed | Initial procedure creation                     | Dr. Ahmed Al-Mansouri |
| 1.3     | September 8, 2022 | Tariq Al-Rasheed | Added storage tiering procedures               | Mohammed Al-Zahrani   |
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## APPROVAL SIGNATURES

| Role                         | Name                 | Signature | Date         |
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| Information Security Officer | Omar Al-Rashid       | [Signed]  | May 19, 2024 |

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## 1.4 Regulatory Alignment

| Regulation           | Requirement                | Phase Applicability |
|----------------------|----------------------------|---------------------|
| NDMO Domain 4        | Data lifecycle controls    | All phases          |
| PDPL Article 18      | Retention limitation       | Phase 4, 5          |
| CBAHI Standards      | Medical record retention   | Phase 2, 4          |
| ZATCA Requirements   | Financial record retention | Phase 2, 4          |
| MOH Circular 2022/31 | Health data preservation   | Phase 4             |

## 2. PHASE 1: DATA CREATION / ACQUISITION

### 2.1 Data Creation Principles

All data created within Company A systems must adhere to the following principles:

| Principle          | Description                                     | Implementation                     |
|--------------------|-------------------------------------------------|------------------------------------|
| Purpose Limitation | Data created only for defined business purposes | Purpose captured in metadata       |
| Data Minimization  | Only necessary data elements collected          | Field-level justification required |
| Accuracy           | Data accurate at point of creation              | Validation rules enforced          |
| Classification     | Security classification assigned at creation    | Auto-classification rules          |

### 2.2 Manual Data Entry Controls

#### 2.2.1 Input Validation Rules

| Validation Type  | Description                                  | Systems      |
|------------------|----------------------------------------------|--------------|
| Data Type Check  | Ensures correct format (integer, text, date) | All systems  |
| Range Validation | Values within acceptable boundaries          | Epic, Oracle |
| Pattern Matching | Format verification (National ID, phone)     | All systems  |



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| Resource Type | Validation             | Source System      |
|---------------|------------------------|--------------------|
| Patient       | Demographics match MPI | External providers |

|                  |                           |                       |
|------------------|---------------------------|-----------------------|
| Encounter        | Valid patient reference   | Referral sources      |
| Observation      | LOINC code validation     | Laboratory interfaces |
| Medication       | RxNorm code validation    | Pharmacy systems      |
| DiagnosticReport | Valid performer reference | Imaging centers       |

### 2.4.2 API Data Acquisition Controls

| Control        | Implementation                    |
|----------------|-----------------------------------|
| Authentication | OAuth 2.0 with client credentials |
| Authorization  | Scope-based access control        |
| Encryption     | TLS 1.3 for data in transit       |
| Rate Limiting  | 1000 requests/minute per client   |
| Logging        | All API calls logged to Splunk    |

## 3. PHASE 2: DATA STORAGE / MAINTENANCE

### 3.1 Storage Tiering Strategy

Company A implements a tiered storage architecture to optimize cost and performance:

#### 3.1.1 Storage Tier Definitions

| Tier    | Storage Type            | Performance     | Cost/TB/Month | Data Age     | Use Case               |
|---------|-------------------------|-----------------|---------------|--------------|------------------------|
| Hot     | NVMe SSD (Pure Storage) | < 1ms latency   | 850 SAR       | 0-12 months  | Active patient records |
| Warm    | SAS HDD (NetApp)        | < 10ms latency  | 280 SAR       | 12-36 months | Historical records     |
| Cold    | Object Storage (S3)     | < 100ms latency | 75 SAR        | 36-84 months | Archive access         |
| Glacier | Tape/Deep Archive       | Hours           | 25 SAR        | 84+ months   | Compliance retention   |

#### 3.1.2 Automatic Data Movement Rules

| Source Tier | Target Tier             | Trigger Condition | Automation Tool   |
|-------------|-------------------------|-------------------|-------------------|
| Hot → Warm  | Last access > 12 months | Automated policy  | NetApp FabricPool |
| Warm → Cold | Last access > 24 months | Automated policy  | Informatica ILM   |

|                |               |               |                     |
|----------------|---------------|---------------|---------------------|
| Cold → Glacier | Age > 7 years | Scheduled job | AWS Lifecycle Rules |
|----------------|---------------|---------------|---------------------|

### 3.1.3 Storage Allocation by System

| System       | Hot Storage | Warm Storage | Cold Storage | Total  |
|--------------|-------------|--------------|--------------|--------|
| Epic EHR     | 12 TB       | 8 TB         | 15 TB        | 35 TB  |
| Philips PACS | 45 TB       | 120 TB       | 280 TB       | 445 TB |
| Oracle EBS   | 4 TB        | 6 TB         | 8 TB         | 18 TB  |
| Oracle HCM   | 500 GB      | 300 GB       | 200 GB       | 1 TB   |
| SharePoint   | 2 TB        | 3 TB         | 5 TB         | 10 TB  |

## 3.2 Database Maintenance Procedures

### 3.2.1 Oracle Database Maintenance Schedule

| Task                     | Frequency | Window         | DBA Responsible        |
|--------------------------|-----------|----------------|------------------------|
| Statistics Gathering     | Daily     | 02:00-04:00    | Automated (DBMS_STATS) |
| Index Rebuild            | Weekly    | Saturday 01:00 | Khalid Al-Mutairi      |
| Tablespace Monitoring    | Daily     | Continuous     | Nagios Alerts          |
| Archive Log Management   | Daily     | Every 4 hours  | Automated              |
| RMAN Backup Verification | Weekly    | Sunday 06:00   | Fahad Al-Dosari        |
| Patch Assessment         | Monthly   | 1st Tuesday    | Tariq Al-Rasheed       |
| Patch Application        | Quarterly | 3rd Saturday   | Full DBA Team          |

### 3.2.2 SQL Server Maintenance Schedule

| Task                   | Frequency    | Window         | DBA Responsible |
|------------------------|--------------|----------------|-----------------|
| Index Defragmentation  | Weekly       | Sunday 02:00   | Automated Job   |
| Statistics Update      | Daily        | 03:00          | Automated Job   |
| Integrity Checks       | Weekly       | Saturday 04:00 | Ahmad Al-Shehri |
| Transaction Log Backup | Every 15 min | Continuous     | Automated       |
| Full Backup            | Daily        | 01:00          | Veeam Agent     |
| Patch Assessment       | Monthly      | 2nd Tuesday    | Ahmad Al-Shehri |

### 3.2.3 PostgreSQL Maintenance Schedule

| Task              | Frequency  | Window             | DBA Responsible  |
|-------------------|------------|--------------------|------------------|
| VACUUM ANALYZE    | Daily      | 03:00              | pg_cron          |
| Reindex           | Monthly    | 1st Sunday         | Yusuf Al-Qahtani |
| pg_basebackup     | Daily      | 02:00              | Automated        |
| WAL Archive       | Continuous | Real-time          | pgBackRest       |
| Extension Updates | Quarterly  | Maintenance window | Yusuf Al-Qahtani |

## 3.3 Data Integrity Maintenance

### 3.3.1 Integrity Check Procedures

| Check Type            | Frequency | Tool                 | Alert Threshold   |
|-----------------------|-----------|----------------------|-------------------|
| Checksum Verification | Weekly    | DBCC CHECKDB / RMAN  | Any corruption    |
| Referential Integrity | Daily     | Custom SQL scripts   | > 0 orphans       |
| Duplicate Detection   | Weekly    | Informatica IDQ      | > 0.1% duplicates |
| Data Type Validation  | Monthly   | Informatica Profiler | > 0.5% invalid    |

### 3.3.2 Integrity Failure Response

Step 1: Alert received in ServiceNow

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Step 2: DBA assesses scope of corruption

↓

Step 3: If isolated: Block-level recovery from backup

If widespread: Point-in-time recovery

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Step 4: Verify recovery success

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Step 5: Root cause analysis

↓

Step 6: Incident report filed

↓

Step 7: Preventive measures implemented

## 3.4 Data Quality Monitoring

### 3.4.1 Continuous DQ Monitoring Rules

| Rule ID | Description                   | Target System | Threshold | Alert           |
|---------|-------------------------------|---------------|-----------|-----------------|
| DQ-001  | Patient DOB completeness      | Epic          | > 99.9%   | Warning < 99.9% |
| DQ-002  | National ID format valid      | Epic          | = 100%    | Critical < 100% |
| DQ-003  | Invoice amount positive       | Oracle EBS    | = 100%    | Critical < 100% |
| DQ-004  | Diagnosis code valid (ICD-10) | Epic          | > 99%     | Warning < 99%   |
| DQ-005  | Employee email unique         | Oracle HCM    | = 100%    | Critical < 100% |

### 3.4.2 DQ Dashboard Metrics

| Metric               | Current Value | Target | Trend |
|----------------------|---------------|--------|-------|
| Overall Completeness | 99.4%         | 99.5%  | ↑     |
| Overall Accuracy     | 98.7%         | 99.0%  | →     |
| Overall Consistency  | 99.1%         | 99.0%  | ↑     |
| Overall Timeliness   | 97.8%         | 98.0%  | ↓     |

## 4. PHASE 3: DATA USE / SHARING

### 4.1 Internal Data Use Controls

#### 4.1.1 Access Control Matrix

| Data Category        | Role                   | Access Level | Justification Required |
|----------------------|------------------------|--------------|------------------------|
| Patient Demographics | Registration Staff     | Read/Write   | No                     |
| Patient Demographics | Clinical Staff         | Read         | No                     |
| Patient Clinical     | Treating Physician     | Read/Write   | No                     |
| Patient Clinical     | Non-treating Physician | Read         | Patient consent        |
| Financial Data       | Finance Staff          | Read/Write   | Job function           |
| Financial Data       | Department Managers    | Read         | Budget review          |
| Employee HR Data     | HR Staff               | Read/Write   | Job function           |
| Employee HR Data     | Line Managers          | Limited Read | Direct reports only    |

### 4.1.2 Data Access Request Workflow

Step 1: User submits Data Access Request in ServiceNow



Step 2: Line Manager approves business need



Step 3: Data Steward verifies data classification



Step 4: Information Security reviews risk



Step 5: Access provisioned via AD Group



Step 6: User acknowledges acceptable use



Step 7: Access logged in Identity Governance system

### 4.1.3 Default Access Principles

| Principle             | Implementation                          |
|-----------------------|-----------------------------------------|
| Least Privilege       | Minimum access for job function         |
| Need-to-Know          | Access limited to required data         |
| Read-Only Default     | Write access requires explicit approval |
| Time-Limited          | Access reviews every 90 days            |
| Segregation of Duties | Conflicting roles prevented             |

## 4.2 External Data Sharing Controls

### 4.2.1 Data Sharing Agreement Requirements

All external data sharing requires a valid Data Sharing Agreement (DSA) covering:

| Element           | Required Content                   |
|-------------------|------------------------------------|
| Parties           | Legal names, addresses, contacts   |
| Purpose           | Specific, documented purpose       |
| Data Elements     | Detailed list of shared fields     |
| Duration          | Maximum 2 years, renewable         |
| Security Controls | Minimum security requirements      |
| Incident Response | Breach notification obligations    |
| Termination       | Data return/destruction procedures |

|            |                                |
|------------|--------------------------------|
| Compliance | PDPL, NDMO, sector regulations |
|------------|--------------------------------|

### 4.2.2 Data Sharing Approval Workflow

| Data Classification | Approval Chain                 |
|---------------------|--------------------------------|
| PUBLIC              | Data Steward                   |
| INTERNAL            | Data Owner + Data Steward      |
| CONFIDENTIAL        | Data Owner + CDO + Legal       |
| SECRET              | Data Owner + CDO + Legal + CEO |

### 4.2.3 Approved Data Sharing Methods

| Method           | Use Case              | Security Controls                 |
|------------------|-----------------------|-----------------------------------|
| SFTP             | Large batch files     | TLS 1.3, IP whitelist, SSH keys   |
| API              | Real-time integration | OAuth 2.0, TLS 1.3, rate limiting |
| Secure Email     | Small files, ad-hoc   | Microsoft Purview encryption      |
| Azure Data Share | Cloud-to-cloud        | Azure AD authentication           |
| Physical Media   | Air-gapped transfer   | Encrypted USB, chain of custody   |

## 4.3 Data Anonymization for Analytics

### 4.3.1 Anonymization Techniques

| Technique            | Description                    | Use Case          |
|----------------------|--------------------------------|-------------------|
| Masking              | Replace with fictitious values | Test environments |
| Tokenization         | Replace with reversible token  | Payment card data |
| Generalization       | Reduce precision               | Age ranges vs DOB |
| Suppression          | Remove sensitive fields        | Public reporting  |
| K-anonymity          | Ensure k-1 identical records   | Research datasets |
| Differential Privacy | Add statistical noise          | Aggregate queries |

### 4.3.2 Anonymization Workflow

Step 1: Analytics team submits data request



Step 2: Privacy Office assesses re-identification risk

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Step 3: Select appropriate anonymization technique

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Step 4: Apply anonymization using Informatica TDM

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Step 5: Validate no direct/indirect identifiers remain

↓

Step 6: Generate anonymization certificate

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Step 7: Release data to analytics environment

## 5. PHASE 4: DATA ARCHIVAL

### 5.1 Archival Criteria

Data transitions to archive status when meeting any of the following criteria:

| Criterion    | Definition                         | Example                      |
|--------------|------------------------------------|------------------------------|
| Age-Based    | Exceeds active retention period    | Patient discharged > 2 years |
| Event-Based  | Business event triggers archive    | Project completed            |
| Access-Based | No access in defined period        | Unaccessed > 18 months       |
| Regulatory   | Meets regulatory archive threshold | Fiscal year close + 1 year   |

### 5.2 Archival Procedure

#### 5.2.1 Automated Archival Workflow

Step 1: Informatica ILM identifies archive candidates

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Step 2: Generate candidate report for Data Owner review

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Step 3: Data Owner approves/exceptions within 14 days

↓

Step 4: Archival job scheduled for maintenance window

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Step 5: Data moved from production tables to archive tables:

TBL\_PATIENTS → ARC\_PATIENTS

TBL\_ENCOUNTERS → ARC\_ENCOUNTERS

TBL\_ORDERS → ARC\_ORDERS

↓

Step 6: Checksum verification confirms data integrity

↓

Step 7: Archive index updated for search capability

↓

Step 8: Production space reclaimed

↓

Step 9: Archival manifest logged for audit

### 5.2.2 Archive Table Naming Convention

| Production Table | Archive Table  | Archive Partition      |
|------------------|----------------|------------------------|
| TBL_PATIENTS     | ARC_PATIENTS   | ARC_PATIENTS_2024_Q1   |
| TBL_ENCOUNTERS   | ARC_ENCOUNTERS | ARC_ENCOUNTERS_2024_Q1 |
| TBL_ORDERS       | ARC_ORDERS     | ARC_ORDERS_2024_Q1     |
| TBL_INVOICES     | ARC_INVOICES   | ARC_INVOICES_FY2024    |

## 5.3 Archive Access Procedures

### 5.3.1 Archive Retrieval Request

| Field                  | Description                           |
|------------------------|---------------------------------------|
| Requester              | Name and department                   |
| Business Justification | Specific need for archived data       |
| Data Elements          | Tables/fields required                |
| Time Period            | Archive date range                    |
| Urgency                | Standard (5 days) / Urgent (24 hours) |
| Target Location        | Where to restore data                 |

### 5.3.2 Archive Retrieval SLA

| Priority      | Definition              | Retrieval SLA   | Approval     |
|---------------|-------------------------|-----------------|--------------|
| P1 - Critical | Legal/regulatory demand | 4 hours         | CDO          |
| P2 - Urgent   | Patient care impact     | 24 hours        | Data Owner   |
| P3 - Standard | Business analysis       | 5 business days | Data Steward |

|              |                      |                  |           |
|--------------|----------------------|------------------|-----------|
| P4 - Routine | Historical reference | 10 business days | Automated |
|--------------|----------------------|------------------|-----------|

## 5.4 Long-Term Preservation

### 5.4.1 Format Migration Schedule

| Data Type    | Current Format | Migration Trigger   | Target Format  |
|--------------|----------------|---------------------|----------------|
| Documents    | PDF/A-1b       | Format obsolescence | PDF/A-3        |
| Images       | DICOM          | Vendor end-of-life  | DICOM (latest) |
| Databases    | Oracle 19c     | Support end         | Oracle 23c     |
| Spreadsheets | XLSX           | Format change       | ODS or CSV     |

### 5.4.2 Preservation Validation

| Check                     | Frequency | Tool              |
|---------------------------|-----------|-------------------|
| File integrity (checksum) | Quarterly | Automated scripts |
| Format readability        | Annually  | Sample validation |
| Metadata completeness     | Quarterly | Informatica EDC   |
| Index accuracy            | Annually  | Reconciliation    |

## 6. PHASE 5: DATA DESTRUCTION

### 6.1 Destruction Criteria

Data is eligible for destruction when ALL of the following conditions are met:

| Condition                  | Verification Method           |
|----------------------------|-------------------------------|
| Retention period expired   | Retention schedule lookup     |
| No active legal hold       | Legal department confirmation |
| No pending audit           | Internal audit clearance      |
| No active investigation    | Security/HR clearance         |
| Business approval obtained | Data Owner sign-off           |

## 6.2 Destruction Methods

### 6.2.1 Logical Destruction (Databases)

| Method         | Command                          | Use Case           | Verification           |
|----------------|----------------------------------|--------------------|------------------------|
| DELETE         | DELETE FROM table WHERE criteria | Individual records | Row count verification |
| TRUNCATE       | TRUNCATE TABLE archive_table     | Full table removal | Table empty check      |
| DROP PARTITION | ALTER TABLE DROP PARTITION       | Time-based removal | Partition list         |
| DROP TABLE     | DROP TABLE archive_table         | Complete removal   | Object dictionary      |

### 6.2.2 Physical Destruction (Media)

| Media Type | Method                   | Standard            | Certificate Required |
|------------|--------------------------|---------------------|----------------------|
| HDD        | Degaussing + Shredding   | NIST SP 800-88      | Yes                  |
| SSD        | Crypto-erase + Shredding | NIST SP 800-88      | Yes                  |
| Tape       | Incineration             | DoD 5220.22-M       | Yes                  |
| Paper      | Cross-cut shredding      | DIN 66399 Level P-4 | No                   |
| Optical    | Shredding                | DIN 66399 Level O-3 | No                   |

### 6.2.3 Destruction Workflow

Step 1: System generates "Eligible for Destruction" report

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Step 2: Data Steward reviews and validates eligibility

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Step 3: Data Owner provides written approval

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Step 4: Legal confirms no hold applies

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Step 5: Destruction scheduled during maintenance window

↓

Step 6: Execute destruction:

- Database: Run approved deletion script
- Files: Secure delete utility
- Media: Vendor pickup for physical destruction

↓

Step 7: Verify destruction complete:

- Database: Confirm zero row count
- Files: Confirm files no longer accessible

- Media: Vendor certificate received

↓

Step 8: Update data catalog (Informatica EDC)

↓

Step 9: Generate Certificate of Destruction

↓

Step 10: Archive certificate for 7 years

## 6.3 Certificate of Destruction

Each destruction event generates a formal certificate containing:

| Element            | Description                            |
|--------------------|----------------------------------------|
| Certificate ID     | Unique identifier (COD-2024-XXXX)      |
| Data Description   | Dataset name, table, system            |
| Record Count       | Number of records destroyed            |
| Date Range         | Data date range covered                |
| Destruction Date   | Date and time of destruction           |
| Method             | Logical or physical destruction method |
| Executor           | Name and title of person executing     |
| Witness            | Name and title of witness              |
| Approval Reference | Data Owner approval reference          |
| Verification       | Confirmation method used               |

## 6.4 Destruction Exceptions

### 6.4.1 Legal Hold Process

Step 1: Legal issues hold notification

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Step 2: Data Steward identifies affected datasets

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Step 3: Destruction jobs suspended for affected data

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Step 4: Hold logged in Legal Hold Register

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Step 5: Quarterly review of active holds

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Step 6: Legal releases hold when matter resolved



Step 7: Data returns to normal lifecycle

### 6.4.2 Active Legal Holds

| Hold ID     | Matter             | Data Scope      | Start Date   | Review Date  |
|-------------|--------------------|-----------------|--------------|--------------|
| LH-2024-001 | Insurance claim    | Encounters 2023 | Jan 15, 2024 | Apr 15, 2024 |
| LH-2024-002 | Employment dispute | HR records 2022 | Feb 28, 2024 | May 28, 2024 |
| LH-2024-003 | Regulatory inquiry | Financial 2023  | Mar 10, 2024 | Jun 10, 2024 |

## 7. SYSTEM-SPECIFIC PROCEDURES

### 7.1 Epic EHR Data Lifecycle

| Phase   | Procedure                                      | Timing                    |
|---------|------------------------------------------------|---------------------------|
| Create  | Patient registration, encounter documentation  | Real-time                 |
| Store   | Epic Chronicles database (Oracle RAC)          | Continuous                |
| Use     | Clinical workflows, reporting                  | Per access policy         |
| Archive | Epic Caboodle (data warehouse archive)         | After discharge + 2 years |
| Destroy | Per CBAHI: Patient lifetime + 10 years minimum | Automated review          |

### 7.2 Oracle EBS Financial Data Lifecycle

| Phase   | Procedure                             | Timing                           |
|---------|---------------------------------------|----------------------------------|
| Create  | Invoice entry, journal posting        | Real-time                        |
| Store   | Oracle EBS database                   | Continuous                       |
| Use     | Financial reporting, AP/AR processing | Per access policy                |
| Archive | Oracle ILM partitioning               | After fiscal year close + 1 year |
| Destroy | Per ZATCA: 10 years minimum           | Automated review                 |

### 7.3 Philips PACS Imaging Data Lifecycle

| Phase   | Procedure                                  | Timing                 |
|---------|--------------------------------------------|------------------------|
| Create  | DICOM acquisition from modalities          | Real-time              |
| Store   | Philips VuePACS (Hot tier)                 | Continuous             |
| Use     | Diagnostic interpretation, clinical review | Per access policy      |
| Archive | Philips Archive (Cold storage)             | After study + 6 months |
| Destroy | Per CBAHI: Patient lifetime + 10 years     | Automated review       |

## 7.4 Microsoft 365 Collaboration Data Lifecycle

| Phase   | Procedure                            | Timing                 |
|---------|--------------------------------------|------------------------|
| Create  | Document creation, email composition | Real-time              |
| Store   | SharePoint Online, Exchange Online   | Continuous             |
| Use     | Collaboration, communication         | Per access policy      |
| Archive | Microsoft Purview retention labels   | Per retention schedule |
| Destroy | Microsoft Purview disposition        | Automated per policy   |

# 8. MONITORING AND COMPLIANCE

## 8.1 Lifecycle Metrics Dashboard

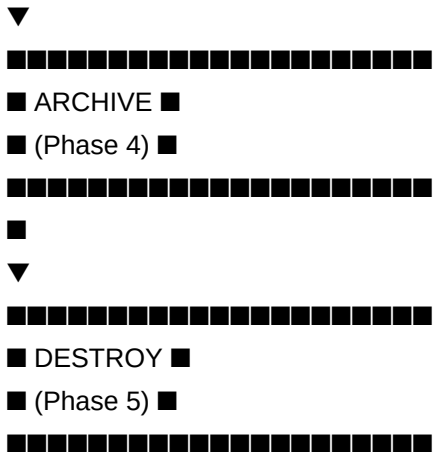
| Metric                | Current | Target   | Status |
|-----------------------|---------|----------|--------|
| Data in Hot Tier      | 68 TB   | < 75 TB  | ✓      |
| Archive Backlog       | 2.3 TB  | < 5 TB   | ✓      |
| Destruction Backlog   | 180 GB  | < 500 GB | ✓      |
| Legal Holds Active    | 3       | N/A      | Info   |
| Failed Lifecycle Jobs | 2       | 0        | ■      |

## 8.2 Compliance Audit Trail

All lifecycle events are logged to the central audit repository:

| Event Type | Logged Data | Retention |
|------------|-------------|-----------|
|------------|-------------|-----------|





## APPENDIX B: RETENTION SCHEDULE SUMMARY

| Data Category           | Active Period          | Archive Period | Total Retention          | Destruction Method |
|-------------------------|------------------------|----------------|--------------------------|--------------------|
| Patient Medical Records | 2 years post-discharge | 8 years        | 10 years minimum         | Logical delete     |
| Patient Imaging (PACS)  | 6 months               | 9.5 years      | 10 years minimum         | Logical delete     |
| Financial Transactions  | 1 year post-close      | 9 years        | 10 years                 | Logical delete     |
| Employee Records        | Employment + 1 year    | 6 years        | 7 years post-termination | Logical delete     |
| Contracts               | Contract term + 1 year | 9 years        | 10 years post-expiry     | Logical delete     |
| Audit Logs              | 1 year                 | 6 years        | 7 years                  | Logical delete     |
| Email (General)         | 3 years                | 4 years        | 7 years                  | Auto-delete        |
| Email (Legal/Executive) | 5 years                | 5 years        | 10 years                 | Manual review      |

Document End

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